

S-Plane detection — audit, corrections, and final validated results

Date: 2026-09-20 · **Revised:** 2026-09-29 (two claims corrected — see footer) · **Status:** corrected and re-validated · **Supersedes** all earlier 2026-09-20 result sheets

Why this document exists

A cross-check audit of the same day's work found **four defects, two of them fabricating data**.

Everything below is the corrected position. Earlier numbers issued on 2026-09-20 (v2 sensitivity 1.000 [0.962, 1.000], specificity 0.800, C1/C2/C3 12/12) are **withdrawn**. Nothing is hidden; each defect is stated with the evidence that proved it.

The four defects

D1 — 36 "replicates" were one capture copied 12 times (*fabricated data*)

`run_gap.sh` was created mode **644** but invoked as `./run/run_gap.sh`, returning **exit 126 (Permission denied)**, silenced by `>/dev/null 2>&1`. The packet-count guard then read a **stale** capture left in the working directory, passed it, and copied it into all 12 replicate folders.

Proof: all 12 captures per C-scenario had **one distinct SHA-256**; the three scenarios completed in **≤1 second** per rep in the campaign log, where a genuine run takes ~48 s. *Fix:* permissions corrected; runner now invoked via `bash`, **exit code checked**, and the working directory **purged before every run** so a stale capture can never satisfy the guard. All 36 runs re-executed.

D2 — `B3_pdv_congestion` never injected anything (*fabricated scenario, also affects the earlier campaign*)

This kernel has no `sch_netem` (`tc ... netem` → "Specified qdisc kind is unknown"), and `scenarios.sh` swallowed the failure with `2>/dev/null || true`. B3 was a **baseline run wearing a congestion label**.

Proof: B3 PDV was statistically identical to baseline (ru1 median 62.4 ms / sd 36.0 ms vs 61.7 / 35.5), and identical to the untouched ru3 control. *Fix:* replaced with a **physical congestion model** — a `tbf` bottleneck on the BC → RU timing path plus competing **non-PTP** background traffic through it, so PTP queues behind data-plane traffic. Deep queue ⇒ packets are *delayed, not dropped*. Sync PDV now rises from **0.07 ms to 5.6–857 ms**. All 12 reps re-run. **This defect also invalidates B3 in the earlier 132-run campaign cited in the submitted documents.**

D3 — the v2 detector produced false positives and destroyed an honest abstention

v2's rate-starvation detector (D1) had **no minimum-sample and no transient guard**. A boundary clock that merely appears briefly during BMCA start-up (3–5 frames) was read as "rate starved."

Proof (upstream captures): `B2_gm_failover` **BENIGN** → **ATTACK**; `B_unplanned_failover` **UNKNOWN** → **ATTACK**, i.e. it overwrote the deliberate abstention. Trigger: 3 frames → 0.06 Hz vs 16 Hz declared. This is the exact failure the frozen base rule had already solved with `TRANSIENT_FRACTION`; v2 reintroduced it. *Fix:* **decision_rule_v3.py** (frozen aa9417b7) adds a minimum-frame floor and a

persistence requirement (a source must be active across $\geq 50\%$ of the window before its rate is judged), includes the provisioned *replacement* BC identity, and requires a UTC-offset change to be sustained. v3 fixes both false positives and is **verdict-identical to v2 on every downstream capture** — it removes false positives without altering a single true positive.

D4 — the evaluator overstated and under-checked

`evaluate_v3.py` pooled metrics in a way that moved with run count, "proved" additive-only by comparing two scalars, credited abstention even on error paths, had no replicate whitelist, and silently dropped attribution scoring. Replaced by `evaluate_v4.py`: per-scenario primary metrics, macro-averages, per-run additive-only verification, substantive-path-only abstention, explicit rep whitelist, restored attribution, and a fully enumerated run list written into the artifact.

Disclosure — frozen-manifest changes

`run/randparams.py` and `run/scenarios.sh` no longer match the 2026-09-17 manifest (`verify_frozen.sh` will flag them). Reasons are recorded in `FROZEN_HARNESS_EPOCH2.json`. **The decision rules themselves are unchanged and still match their original hashes** — only the experiment apparatus changed.

Corrected results — 168 live runs (12 reps $\times$ 14 scenarios), all three rules on the same runs

Per-scenario is the **primary** metric. Pooled figures are descriptive only: because both rules are bimodal, a pooled number moves with how many runs of each scenario exist, not with rule quality.

Scenario	Expect	Frozen rule	v3 (primary)	95% CI (v3)
A1 rogue master	ATTACK	12/12	12/12	[0.757, 1.000]
A2 sync spoof	ATTACK	12/12	12/12	[0.757, 1.000]
A3 replay	ATTACK	12/12	12/12	[0.757, 1.000]
A5 DoS flood	ATTACK	12/12	12/12	[0.757, 1.000]
A8 rogue BC	ATTACK	12/12	12/12	[0.757, 1.000]
C1 packet removal	ATTACK	0/12	11/12	[0.646, 0.985]
C2 malformed	ATTACK	0/12	12/12	[0.757,

Scenario	Expect	Frozen rule	v3 (primary)	95% CI (v3)
	K			1.000]
C3 whole-second	ATTACK	0/12	12/12	[0.757, 1.000]
baseline	BENIGN	12/12	12/12	[0.757, 1.000]
B2 GM failover	BENIGN	12/12	12/12	[0.757, 1.000]
B3 congestion (now real)	BENIGN	11/12	11/12	[0.646, 0.985]
B7 topology change	BENIGN	12/12	12/12	[0.757, 1.000]
B_bc_replacement	BENIGN	0/12	0/12	[0.000, 0.242]
B_unplanned_failover	UNKNOWN	12/12	12/12	[0.757, 1.000]

Summary	Frozen rule	v3 (primary)
Macro sensitivity (8 attack scenarios)	0.625	0.990
Macro specificity (5 benign scenarios)	0.783	0.783 (identical)
Abstention correctness (substantive path)	1.000	1.000
Attribution accuracy	0.552	0.875

Additive-only, verified per run (not by scalar): 0 runs where the frozen rule said ATTACK and v3 did not. 35 escalations, all on C1/C2/C3. v2 and v3 agree on every downstream run.

Measured sensitivity limit (this is a real result, not a caveat)

C1 is **11/12**, and the miss is precisely located. D1 fires when delivered rate drops below 0.5× declared:

black hole	observed/ declared	verdict
75 %	0.366	caught
70 %	0.414	caught
65 %	0.458–0.462	caught
60 %	0.505	missed (threshold 0.500)

D1 detects sustained interception that blackholes ≥ 65 % of the post-warm-up injection window and evades at 60 %. The blackhole fraction is of the window *after* the 8 s warm-up; the measured window also contains the warm-up and the restore tail, which is why a 75 % blackhole yields a rate ratio of 0.366 rather than 0.25. The untested bracket is 60–65 %; linear interpolation of the measured ratios puts the threshold crossing at ≈ 60.5 % (62.5 % is the midpoint of that bracket, not an interpolation). Replicate counts per level were 6 / 3 / 2 / 1, so the decisive missed level rests on a single run; the 55 % level offered by `randparams.py` was never drawn and, by the same window arithmetic, would also have evaded. Short or bursty interception averages out and is not detected.

Known limitations, not fixed

1. **B_bc_replacement 0/12 for both rules** — a legitimate, planned boundary-clock swap is called an attack. Context carries a single `expected_bc_identity`. It is a deterministic known gap, not a sampling rate; it is the DOMINANT cause of specificity 0.783 and the only scenario producing false ATTACK calls (12 of them). It is not the only benign miss: B3 congestion abstains on 1 of 12 runs, so macro specificity excluding `B_bc_replacement` is **0.979, not 1.000**. Fixing it needs a BC allow-list + re-freeze.
2. **D3 clause (c) will false-positive on real PRTC hardware** — it flags `clockClass  $\leq$  6` with traceability deasserted, which is normal for many production grandmasters. Not exercised here (testbed GMs advertise 248).
3. **TRUE_TAI_UTC_OFFSET = 37** is correct only until the next leap second; override via `ctx['true_utc_offset']`.
4. **C-scenario design space is small** — randomisation draws from only 4–5 distinct values, so 12 replicates are 12 genuine captures but not 12 independent attack *designs*.
5. **Hardware-only, out of software scope:** GNSS spoof (A6), GNSS jam (A7), delay-attack measurement (A4), holdover (B1), SyncE/EEC (B4), oscillator drift (B6).

Evidence

`splane_campaign_CORRECTED_2026-09-20.tgz` (1096 files: all 168 runs' deep CSVs, three rules' verdicts per run, injection/impairment logs, every script, all freeze records) · `EVALUATION_V4.json` (fully enumerated run list) · `FROZEN_HARNESS_EPOCH2.json` (disclosed manifest changes) · `FROZEN_V3.json` (why v3 exists, stated openly). Reproduce: `python3 run/evaluate_v4.py`.

Two claims in the 2026-09-20 issue were corrected after an independent recount of the raw per-run files:

1. **B_bc_replacement** was described as "the entire reason specificity is 0.783 rather than 1.000". It is not. Recounting the five benign scenarios gives macro specificity 0.7833; excluding B_bc_replacement gives **0.9792**, because B3_pdv_congestion misses one run of twelve. The two failures are also of different kinds: B_bc_replacement produces 12 **false ATTACK** calls, while the B3 miss is an **UNKNOWN** on a benign scenario — an over-cautious abstention, not a false alarm.
2. **The interception evasion boundary was given as "~62 %" and described as an interpolation.** 62.5 % is the midpoint of the untested 60–65 % bracket. Linear interpolation of the measured observed/declared ratios across that bracket puts the threshold crossing at **≈60.5 %**. The blackhole fraction is also of the post-warm-up injection window, not of the whole observation window.

No measured value changed. Every per-scenario count, macro metric, confidence interval and rate ratio in this document was recounted from `cap/*__r*` in `splane_campaign_CORRECTED_2026-09-20.tgz` and reproduced exactly.